



Natural Language Processing Applications



BITS Pilani
Pilani Campus

Dr. Chetana Gavankar, Ph.D,
IIT Bombay-Monash University Australia
Chetana.gavankar@pilani.bits-pilani.ac.in



Session 5: Conversational AI

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AI Assistants



Notification Assistant



Hi there - just a friendly reminder that your insurance policy expires in a month. Make sure to renew it in our member portal.

FAQ Assistant

I need to renew my renters insurance. How much will it be?



You can calculate your renewal price on our website here: xyz.com/renew

Contextual Assistant

I need to renew my renters insurance. How much will it be?



I'd be happy to check for you. Firstly, are you still living in the same apartment?

Yes



Great - so just confirming it's 980 sq ft?

Yes



Thanks! Your new rate from September 1st onwards would be \$10 / month.



Would you like me to renew your policy for you right now?

Sure



Great. I've sent you a confirmation to your email.

AI Assistants



Personalized Assistant

- Assistant knows you much more in detail
- Quickly checks a few final things before giving you a quote tailored to your actual situation.



I can see your details are almost the same, except now you might want coverage for your new laptop. Additional coverage is only \$4 a month more for full coverage. Sound ok?

Sounds good!



Autonomous Organization of Assistants

- Group of AI assistants that know every customer personally
- Eventually run large parts of company operations—from lead generation over marketing, sales, HR, or finance



Conversational AI Approaches and real world applications



Approach	Short Summary	Real-World Applications
Rule-Based	Fixed rules, decision trees, predictable responses	• IVR phone menus • Banking/telecom FAQ bots • HR onboarding flows • Simple customer support chat
IR-Based	Retrieves best match from a knowledge base using keywords or vector search	• Enterprise FAQ search • E-commerce support • Airline & telecom chatbots • Government info portals
Deep Learning (Pre-LLM)	Seq2Seq / early Transformer models that generate basic responses	• Smart reply suggestions • Domain-specific assistants • Simple virtual assistants
LLM-Based	Large pretrained models with reasoning, context understanding, and generative ability	• Advanced chatbots • Coding assistants • Legal/medical Q&A (supervised) • Enterprise RAG systems
Agentic AI	LLM + tools + memory → autonomous task execution and multi-step planning	• Travel booking agents • Financial analysis bots • Customer service copilots • DevOps/coding agents • Workflow automation

QA versus conversational AI



Aspect	Question Answering (QA)	Conversational AI
Definition	A system that answers specific user questions.	A system that engages in a multi-turn dialogue with users.
Scope	Narrow: Answering questions (often factual).	Broad: Dialogue, intent detection, emotion, context, etc.
Example Task	“What is the capital of France?” → “Paris”	“I’m planning a trip.” → “Great! Where are you going?”

Conversational Agents

AKA Dialog Agents



- Phone-based Personal Assistants
 - SIRI (Apple), Alexa (Amazon),
 - Cortana (Microsoft), Google Assistant
- Talking to your car, Pay bills...
- Sales, Marketing, Insurance....
- Clinical uses for mental health, Nurses, Dr. Bot
- Lawyer bots
- Chatting for fun, Mr. FriendBot

Context and Conversation in Virtual Assistants like Siri



- Coreference helps resolve ambiguities
 - U: “Book a table at Il Fornaio at 7:00 with **my mom**”
 - U: “Also send **her** an email reminder”
- Clarification questions:
 - U: “Chicago pizza”
 - S: “Did you mean pizza restaurants in Chicago or Chicago--style pizza?”

Conversation Characteristics



Turns

- dialogue is a sequence of turns
- C1: I need to travel in May.
A1: And, what day in May did you want to travel?
C2: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A2: And you're flying into what city? C3: Seattle.

Speech Acts of Dialog Acts:

- Constatives: answering, claiming, confirming, denying, disagreeing, stating)
- Directives: to do something (advising, asking, forbidding, inviting, ordering, requesting)
- Commissives: committing the speaker to some future course of action (promising, planning, vowing, betting, opposing)
- Acknowledgments: apologizing, greeting, thanking, accepting an acknowledgment

Challenges of Dialog Systems



Grounding:

- acknowledging that the hearer has understood the speaker

Dialog Structure:

- QUESTIONS set up an expectation for an ANSWER. PROPOSALS are followed by ACCEPTANCE (or REJECTION). COMPLIMENTS (“Nice jacket!”) often give rise to adjacency pair DOWNPLAYERS (“Oh, this old thing?”). These pairs, called adjacency pairs are composed of a first pair part and a second pair part

Clarification/correction question

- Which can form a sub dialogue between a REQUEST and a RESPONSE
- Ex: User: What do you have going to UNKNOWN WORD on the 5th?
System: Let’s see, going where on the 5th?

Inference:

- A1: And, what day in May did you want to travel? C2: OK uh I need to be there for a meeting that’s from the 12th to the 15th. Notice that the client does not in fact answer the agent’s question. The client merely mentions a meeting at a certain time

Conversational AI journey

ELIZA (1960s): pattern matching.

1990s–2000s: statistical dialogue models.

2010s: deep learning & seq2seq.

2020+: transformers & LLMs, Agentic AI

Architectures



❑ Rule-based

- Pattern-action rules (Eliza)

❑ Corpus-based

- Information Retrieval
- Neural network encoder-decoder
- Transformers and LLM
- Agentic AI

Eliza: Weizenbaum (1966)



Dialog in Green are by Eliza system.

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

...

WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU?

My father

YOUR FATHER

- Chatbot for delivering Cognitive Behavior Therapy via brief daily conversations
 - Little decision tree of language and menu responses

"What's going on in your world right now?"

"How are you feeling?"

In response to endorsed loneliness:

"I'm so sorry you're feeling lonely. I guess we all feel a little lonely sometimes"

- In a 2-week experiment seemed to reduce depression, but not anxiety

Fitzpatrick, Darcy, Vierhile, 2017. Delivering Cognitive Behavior Therapy to Young Adults with Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial. JMIR Ment Health 4:2.

Simple Chatbot – Rule based Demo

- [NLTK](#) has a module, [nltk.chat](#), which simplifies building Chatbots
- **Chat**: This is a **class** that has all the logic that is used by the chatbot.
- **Reflections**: This is a **dictionary that contains a set of input values and its corresponding output values**. It is an optional dictionary that you can use. You can also create your own dictionary in the same format as below and use it in your code.

IR-based chatbots

- Idea: Mine conversations of human chats or human-machine chats
 - Microblogs: Twitter or Weibo
 - Movie dialogs
- Cleverbot (Carpenter 2017
<http://www.cleverbot.com>)

A Cleverbot conversation

User:	Do you like Doctor Who?
Cleverbot:	Yes, so funny
User:	What's your favorite show?
Cleverbot:	I don't have one
User:	What's your name?
Cleverbot:	Jason I already told you.
User:	No your real name.
Cleverbot:	Stephanie IS my real name.
User:	I thought it was Jason?
Cleverbot:	How did you come to that conclusion.
User:	Because you said so.
Cleverbot:	I have to do my homework.

Two IR-based chatbot architectures

1. Return the response to the most similar turn
 - Take user's turn (q) and find a (tf-idf) similar turn t in the corpus C
 $q = \text{"do you like Doctor Who"}$
 $t' = \text{"do you like Doctor Strangelove"}$
 - Grab whatever the response was to t .

$$r = \text{response} \left(\underset{t \in C}{\operatorname{argmax}} \frac{q^T t}{||q|| ||t||} \right)$$

Yes, so funny

2. Return the most similar turn

$$r = \underset{t \in C}{\operatorname{argmax}} \frac{q^T t}{||q|| ||t||}$$

Do you like Doctor Strangelove

IR-based models of chatbots

- Also fine to use other features like user features, or prior turns
- Or non-dialogue text
 - COBOT chatbot (Isbell et al., 2000)
 - sentences from the Unabomber Manifesto by Theodore Kaczynski, articles on alien abduction, the scripts of “The Big Lebowski” and “Planet of the Apes”.
 - Wikipedia text

IR based Chatbot Demo

- Web scraping using Python
- Retrieve Web page on Chatbots
- Bot answers questions related to the Web page content.

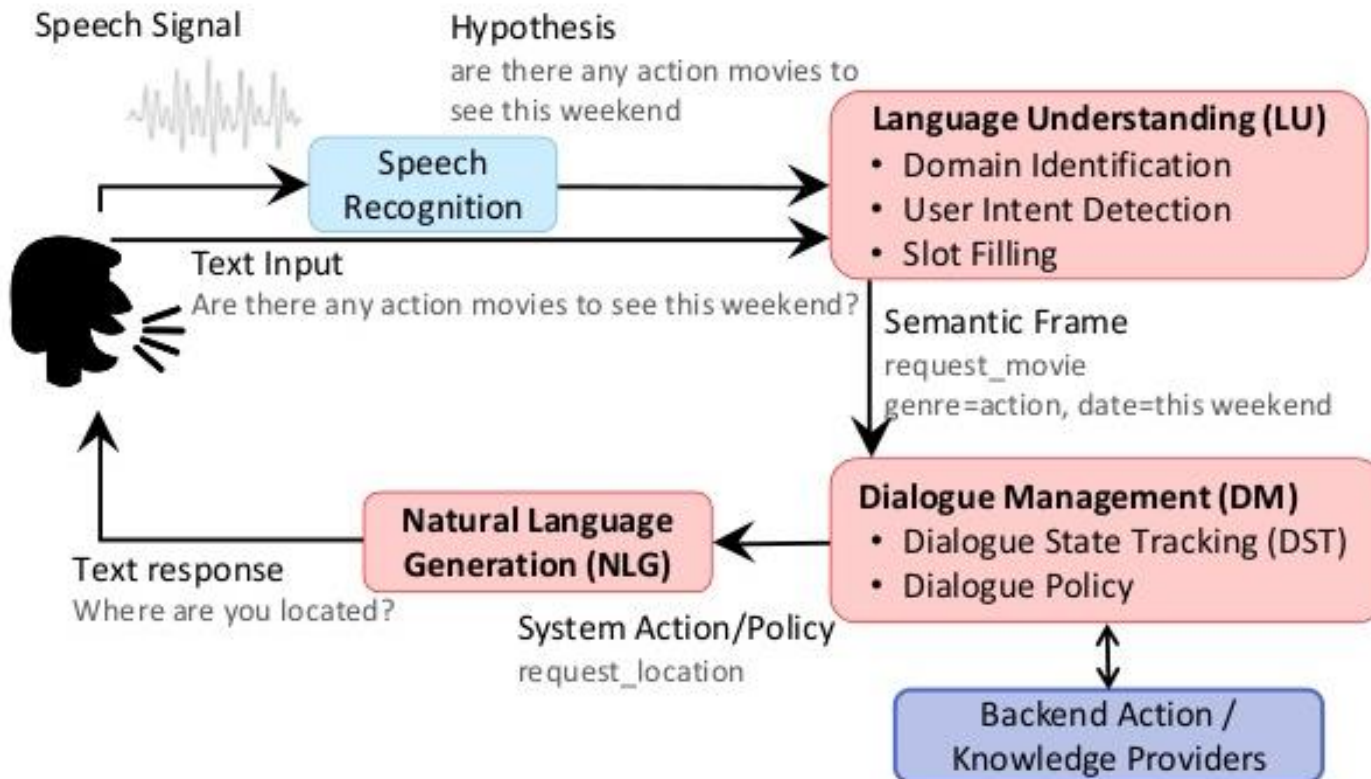
Components of Conversational AI systems



Task-Oriented Dialogue System (Young, 2000)

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<http://rsta.royalsocietypublishing.org/content/358/1769/1389.short>



Components of Conversational AI systems



Building blocks of systems that understand and generate natural language

Natural Language Understanding (NLU)	Dialogue Management	Natural Language Generation (NLG)
Intent classification & slot filling	Conversation state tracking	Template-based generation
Named Entity Recognition (NER)	Rule-based & neural policy	Neural text generation
Semantic parsing	Context window management	Persona & tone consistency
Coreference resolution	Multi-turn coherence	Fluency & coherence scoring
Sentiment & tone detection	Clarification strategies	Multi-modal output support

Hybrid Approach: Combines rule-based precision with neural flexibility for robust, enterprise-ready conversational systems

Approaches for different components of conversational AI system



Approach	NLU (Understand Intent + Extract Slots)	NLG (Generate Responses)	Dialog Management (Planning Flow)	Example Applications
Rule-Based (Finite State / Pattern Matching)	Manual rules, pattern matching (if-else, regex). No learning.	Template-based responses; no variation; deterministic.	Hard-coded flowcharts; state machines; rigid paths.	IVR systems, early chatbots (ELIZA), FAQ bots, customer service scripts.
Statistical NLP (Pre-DL, ML Models)	Machine learning classifiers (SVM, CRF, HMM) for intent & slot tagging; requires labelled data.	Template-based with statistical ranking. Limited generation.	Probabilistic dialog policies (POMDPs).	Early Dialogflow, IBM Watson, telecom/banking bot NLU engines.
Deep Learning (RNNs, LSTMs, Seq2Seq)	Neural intent classifiers; BiLSTM-CRF slot tagging; moderate generalization; still domain-specific.	Seq2Seq or encoder-decoder models; limited creativity; domain-tuned.	Policy networks, trained with supervised or reinforcement learning; still domain-specific.	Alexa Skills (early versions), Rasa DL, Siri NLU pipelines.

Approaches for different components of conversational AI system



Approach	NLU (Understand Intent + Extract Slots)	NLG (Generate Responses)	Dialog Management (Planning Flow)	Example Applications
LLM-Based Conversational AI (GPT, Claude, Gemini)	Intent + slots extracted implicitly using prompts/in-context learning; no task-specific training needed; works zero-shot.	Free-form, high-quality generation; style control; multi-modal; creative.	LLM predicts next system actions; can manage context dynamically; no fixed flow.	ChatGPT-style assistants, enterprise Q&A, customer support bots.
Agentic AI (ReAct, Tool Use, Function Calling, Frameworks)	LLM extracts intents/slots in structured format (JSON, schema) for tools; works zero/low-shot; highly robust.	Multi-step reasoning, tool-calling, API execution, planning. NLG adapts after actions.	Autonomous dialog planning using reasoning loops (Thought → Action → Observation → Response) ; dynamic workflows; memory.	AI agents, workflow automation bots, enterprise assistants, travel/finance/HR multi-step task assistants.

Large Language Models (LLMs) based Conversational AI



Foundation models powering modern conversational AI

Transformer Architecture	Pre-training Strategies	Prompt Engineering	Fine-tuning Methods
Self-attention mechanism	Causal language modeling	Zero-shot & few-shot	Full parameter fine-tuning
Multi-head attention	Masked LM (BERT-style)	Chain-of-Thought (CoT)	LoRA & QLoRA (PEFT)
Positional encoding	RLHF & DPO alignment	Tree-of-Thought (ToT)	Adapter layers
Feed-forward networks	Instruction tuning (SFT)	ReAct prompting	Supervised Fine-Tuning (SFT)
Layer normalization	Constitutional AI (CAI)	System / user / assistant roles	Direct Preference Optimization

Popular LLMs for Conversational AI

GPT-4o / GPT-4	~1T (est)	Reasoning, multimodal, instruction following
Claude 3.5 / 4	~100B+	Long context, safety, nuanced dialogue
Gemini Ultra/Pro	~1T (est)	Multimodal, Google ecosystem
LLaMA 3.x	8B–405B	Open-source, fine-tuning friendly, on-prem
Mistral / Mixtral	7B–56B	Efficient MoE, multilingual support

Retrieval-Augmented Generation (RAG) for Conversational AI



Grounding LLM responses in verified, up-to-date knowledge sources



Retrieval Strategies
Dense retrieval (bi-encoder)
Sparse retrieval (BM25, TF-IDF)
Hybrid = dense + sparse fusion
Re-ranking with cross-encoders
Contextual compression
Parent-child chunking

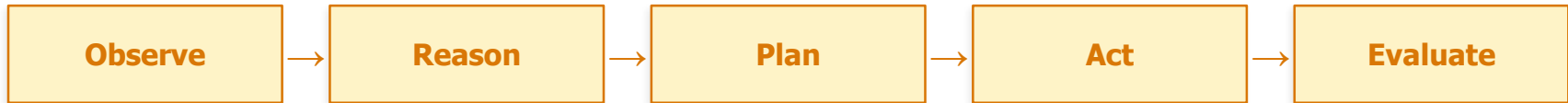
Vector Databases
Pinecone — managed, scalable
Weaviate — hybrid search native
Chroma — lightweight, local dev
pgvector — PostgreSQL extension
Qdrant — Rust-based, fast
FAISS — Meta, in-memory

Advanced RAG Patterns
HyDE (hypothetical doc embeddings)
RAPTOR (tree-based recursive)
Self-RAG with reflection tokens
Graph RAG for structured data
Agentic RAG with tool calling
Multi-hop reasoning chains

Agentic AI for Conversational AI Systems



Autonomous AI agents that reason, plan, and act to complete complex goals



Tool Use & Function Calling

- API calls, web search, code execution
- Calculator, database lookup, file I/O
- Structured output parsing & validation
- MCP (Model Context Protocol) integration

Memory Systems

- Short-term: in-context window
- Long-term: vector DB memory stores
- Episodic & semantic memory patterns
- Session persistence across turns

Multi-Agent Orchestration

- Supervisor → worker agent hierarchy
- Parallel task execution & aggregation
- LangGraph, AutoGen, CrewAI, Swarm
- Error recovery & retry logic

Planning & Reasoning

- ReAct: Reasoning + Acting loop
- Tree-of-Thought deliberate search
- MCTS-based decision planning
- Reflection & self-correction



Hybrid Conversational AI Architecture

Combining LLM, RAG, and Agentic AI into a unified, production-grade system

User Interface Layer

Chat UI · Voice Assistant · REST API · SDK · Omnichannel

Orchestration Layer

LangChain · LangGraph · LlamaIndex · Semantic Kernel · Custom Router

Hybrid Reasoning Core

LLM (GPT-4o / Claude / Gemini) + RAG Pipeline + Agentic Planner

Memory & Context

Short-term (context window) · Long-term (vector DB) · Session state store

Tools & Integrations

Web Search · Databases · External APIs · Code Execution · File I/O

Knowledge & Data Layer

Vector Store · Document DB · Graph DB · Structured Data Sources

Hybrid advantage: Structured precision + neural flexibility + retrieval accuracy + autonomous planning = robust enterprise deployments

Comparison of approaches



Approach	Architecture	Pros	Cons	Best Used For / Applications
1. Rule-Based	Decision trees, flowcharts	- Easy to implement- Predictable behavior	- Hard-coded- Not scalable- Poor with ambiguity	- Simple FAQs- IVR systems- Basic customer service
2. Slot-Filling (Modular)	NLU → DM → NLG	- Structured- Good for well-defined tasks	- Struggles with context- Needs labeled data	- Food ordering bots- Travel booking assistants
3. End-to-End Neural	Seq2Seq / Transformer	- Learns dialogue patterns- More fluid conversations	- Requires lots of data- Poor control/explainability	- Open-domain chat- Virtual companions- Dynamic support
4. Retrieval-Based	Query + Similarity Match → Predefined Reply	- Fast- Accurate with proper data- Grounded	- Limited to existing replies- Low flexibility	- Customer support- Helpdesk bots- FAQ assistants
5. LLM-Based	Prompt → Foundation Model (e.g., GPT)	- Flexible- Handles multiple tasks- Few-shot capable	- Costly- May hallucinate- Lacks structure	- Virtual agents- General assistant bots- Support chatbots
6. Hybrid (LLM + Rules/APIs)	LLM + Dialogue Manager + Tools/APIs	- Best of both worlds- Robust and natural	- Complex to design- Needs integration effort	- Enterprise assistants- Multistep workflows- Copilots



Evaluation of Conversational AI Systems

Systematic assessment of accuracy, safety, and user experience

RAG Evaluation (RAGAS)
Faithfulness
Answer Relevancy
Context Precision
Context Recall
LLM Quality Metrics
BLEU / ROUGE scores
BERTScore (semantic sim.)
Perplexity & coherence
Hallucination rate
Conversational Metrics
Turn-level coherence
Task success rate
User satisfaction (CSAT)
Dialogue act accuracy

Evaluation Tools & Frameworks

RAGAS
End-to-end RAG pipeline evaluation with LLM-as-judge scoring
LangSmith
Tracing, debugging and evaluation for LangChain apps
TruLens
Feedback functions and RAG quality scoring (TruEra)
Promptfoo
Automated red-teaming and prompt regression testing
OpenAI Evals
Model-graded evals, classification, string match
Human Eval
Likert-scale ratings, A/B testing, expert annotation

Evaluation

- Natural Language Understanding component Evaluation:
 1. Slot Error Rate for a Sentence =
$$\frac{\text{\# of inserted/deleted/substituted slots}}{\text{\# of total reference slots for sentence}}$$

Slot tagging is most important since it contributes maximum to quality of Dialog System
 2. Intent Accuracy: Correct intent identified
- Dialog Manager Component Evaluation
 1. State tracking Accuracy
 2. End-to-end evaluation (Task Success)

Evaluation Metrics

“Make an appointment with Chris at 10:30 in Gates 104”

Slot	Filler
PERSON	Chris
TIME	11:30 a.m.
ROOM	Gates 104

Slot error rate: 1/3

Task success: At end, was the correct meeting added to the calendar?

Dialog System Design: User-centered Design



1. Study the user and task
2. Iteratively test the design on users



Ethical Issues in Dialog System Design

- Machine learning systems replicate biases that occurred in the training data.
- Microsoft's Tay chatbot
 - Went live on Twitter in 2016
 - Taken offline 16 hours later
- In that time it had started posting racial slurs, conspiracy theories, and personal attacks
 - Learned from user interactions (Neff and Nagy 2016)

Ethical Issues in Dialog System Design

- Machine learning systems replicate biases that occurred in the training data.
- Dialog datasets
 - Henderson et al. (2017) examined standard datasets (Twitter, Reddit, movie dialogs)
 - Found examples of hate speech, offensive language, and bias
 - Both in the original training data, and in the output of chatbots trained on the data.
- Fairness in Machine Learning
- Dialog agents overwhelmingly given female names, perpetuating female servant stereotype(Paolino, 2017).

Ethical Issues in Dialog System Design: Privacy



- Remember this was noticed in the days of Weizenbaum
- Agents may record sensitive data
 - (e.g. “Computer, turn on the lights [answers the phone –Hi, yes, my password is...”],
- Which may then be used to train a seq2seq conversational model.



Production Environment Tech Stack

Enterprise-grade infrastructure for deploying, monitoring, and scaling conversational AI

Orchestration	Deployment	Observability	Security & Guardrails
LangChain / LangGraph	Azure OpenAI Service	LangSmith tracing	NVIDIA NeMo Guardrails
LlamaIndex	AWS Bedrock	Weights & Biases	Guardrails AI
Haystack (deepset)	Google Vertex AI	Prometheus + Grafana	Lakera / Prompt Shield
Semantic Kernel (MS)	Hugging Face Endpoints	Datadog APM	Azure Content Safety
CrewAI / AutoGen	vLLM (self-hosted)	Arize AI / Phoenix	Presidio (PII detection)

MLOps & CI/CD: Model versioning (DVC / MLflow) · Automated eval gates · Blue-green deployments · Feature flags · A/B testing · Rollback policies

Scaling: Auto-scaling (K8s HPA) · Semantic caching (Redis) · Rate limiting · Load balancing · Streaming (SSE / WebSocket)

Summary

01

Hybrid systems (LLM + RAG + Agentic AI) deliver superior accuracy, groundedness, and task completion vs. any single approach alone.

02

RAG is essential for knowledge-intensive domains — it eliminates hallucinations and keeps answers current without costly model retraining.

03

Agentic patterns (ReAct, tool use, multi-agent orchestration) unlock complex multi-step workflows and seamless enterprise system integration.

04

Rigorous evaluation (RAGAS, LangSmith, human eval) must be embedded in CI/CD pipelines as automated gates before any production deployment.

05

Production readiness demands observability, guardrails, MLOps automation, and scalable cloud-native infrastructure from day one.

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